

Reading

PREPARING TO READ

A BUILD VOCABULARY The words in **blue** are used in the reading passage. Read the text below. Then write the correct form of each word next to its definition.

Scientists around the world are studying insect behavior in order to create tiny robots with the **capabilities** of these amazing creatures. Many insects, for example, are able to land on and take off from tiny surfaces. This could have many applications: military robots, for instance, could enter enemy territory without being seen, while emergency robots could access areas humans and other robots can't.

To better **imitate** insect flight, some engineers are developing ornithopters—small aircraft which get most of their lift from flapping wings. This technology offers several advantages: ornithopters can move in ways regular drones can't, plus they more **realistically** mimic birds and insects—a feature that could be useful to both the military and wildlife researchers.



Engineers are also exploring ways for their flying robots to master **synchronized** flight. This would allow the robots to operate in **swarms**, the way locusts and other insects do. While the applications for this aren't as obvious, it is hypothesized that in some situations, many small robots working together could collectively do much more than one large but **solitary** robot.

The **complexity** of insect and swarm behavior is not to be underestimated—it is the result of thousands of years of evolution—but if we can master it, the benefits would no doubt be immense.

1. _____ (v) to copy or mimic
2. _____ (n) a skill or quality
3. _____ (n) a large group of small animals (usually insects)
4. _____ (n) how complicated something is
5. _____ (adv) in a way that is believable
6. _____ (adj) single; lone
7. _____ (adj) happening at the same time in a well-orchestrated way

B BUILD VOCABULARY Complete the sentences below with the correct form of the words in **blue**. Use a dictionary to help you.

accomplish (v)	anonymous (adj)	coordinate (v)	detect (v)
phenomenon (n)	shrewd (adj)	simulation (n)	uncertainty (n)

1. We might both end up doing the same tasks if we don't _____ our efforts.
2. In order to determine whether or not the robots would be able to handle a real emergency, the engineers first ran them through a series of _____.
3. They could have _____ their goal if they had worked more closely together as a team.
4. The researchers would not have noticed the gas leak if the sensors in the laboratory hadn't _____ it.
5. You have to be careful when doing business with her. She's a really _____ negotiator.
6. They're guilty of groupthink, a(n) _____ that occurs when members of a group simply agree with each other without critically assessing their decisions or other alternatives.
7. Even though she donated the most money to the cause, she chose to remain _____.
8. It was hard to make a sound decision at the time. There was a lot of _____ surrounding the issue.

C USE VOCABULARY Discuss these questions with a partner.

1. Think of a team you're on. What are the strengths and **capabilities** of its various members? Do you feel the team is well balanced?
2. What are some of the **complexities** of working with others in teams?
3. What are some tasks that you could not have **accomplished** without the help of a team? Why couldn't you do the tasks alone?
4. What happens when you fail to **coordinate** with others properly on a project?

D PREDICT You are going to read about swarm behavior. Look at the photos and captions in the reading passage. Then read the first and last paragraphs (A and V). Discuss the questions below with a partner.

Critical Thinking

1. What animals might the passage discuss?
2. What aspects of their behavior might the passage discuss?
3. What human activities or inventions might the passage discuss?
4. What do you think is the main purpose of the article?

Check your answers as you read the passage.

A large flock of starlings flying in a circular pattern over a lake at sunset. The birds are silhouetted against a bright orange and yellow sky, forming a dense, swirling shape. The lake below is calm, reflecting the light from the sky. In the foreground, there are reeds or tall grasses. In the background, there are mountains and some distant lights.

The Smart Swarm

By Peter Miller

Starlings demonstrating swarm behavior over a lake in Wales

The study of swarms is providing insights that can help humans manage complex systems.

A  How do the simple actions of individuals add up to the complex behavior of a group? How do hundreds of honeybees make a critical decision about their hive if many of them disagree? What enables a school of herring to **coordinate** its movements so precisely it can change direction in a flash—like a single, silvery organism? The answer has to do with a remarkable **phenomenon** I call the smart **swarm**.

B A smart swarm is a group of individuals who respond to one another and to their environment in ways that give them the power, as a group, to cope with **uncertainty**, **complexity**, and change. Take birds, for example. There's a small park near the White House in Washington, D.C., where I like to watch flocks of pigeons swirl over the traffic and trees. Sooner or later, the birds come to rest on ledges of buildings surrounding the park. Then something disrupts them, and they're off again in **synchronized** flight.

C The birds don't have a leader. No pigeon is telling the others what to do. Instead, they're each paying close attention to the pigeons next to them, each bird following simple rules as they wheel across the sky. These rules add up to a kind of swarm intelligence—one that has to do with precisely coordinating movement.

D Craig Reynolds, a computer graphics researcher, was curious about what these rules might be. So, in 1986, he created a deceptively simple steering program called boids. In this **simulation**, generic birdlike objects, or boids, were each given three instructions: (1) avoid crowding nearby boids, (2) fly in the average direction of nearby boids, and (3) stay close to nearby

boids. The result, when set in motion on a computer screen, was a convincing simulation of flocking,¹ including lifelike and unpredictable movements.

E At the time, Reynolds was looking for ways to depict animals **realistically** in TV shows and movies. (*Batman Returns* in 1992 was the first movie to use his approach, portraying a swarm of bats and an army of penguins.) He later went on to work at Sony, doing research for games, such as for an algorithm² that simulated in real time as many as 15,000 interacting birds, fish, or people.

F By demonstrating the power of self-organizing models to mimic swarm behavior, Reynolds was also blazing the trail for robotics engineers. A team of robots that could coordinate its actions like a flock of birds could offer significant advantages over a **solitary** robot. Spread out over a large area, a group could function as a powerful mobile sensor net, gathering information about what's out there. If the group encountered something unexpected, it could adjust and respond quickly, even if the robots in the group weren't very sophisticated—just as ants are able to come up with various options by trial and error. If one member of the group were to break down, others could take its place. And, most important, control of the group could be decentralized, not dependent on a leader.

G “In biology, if you look at groups with large numbers, there are very few examples where you have a central agent,” says Vijay Kumar, a professor of mechanical engineering at the University of Pennsylvania. “Everything is very distributed: They don't all talk to each

¹ When animals (usually birds) **flock**, they congregate and do things as a large group.

² An **algorithm** is a mathematical or logical rule which determines how a program or process operates.

other. They act on local information. And they're all **anonymous**. I don't care who moves the chair, as long as somebody moves the chair. To go from one robot to multiple robots, you need all three of those ideas."

H In the future, Kumar hopes to put a networked team of robotic vehicles in the field. One purpose might be as first responders. "Let's say there's a 911 call," he says. "The fire alarm goes off. You don't want humans to respond. You want machines to respond, to tell you what's happening. Before you send firemen into a burning building, why not send in a group of robots?"

I Taking this idea one step further, computer scientist Marco Dorigo's group in Brussels is leading a European effort to create a "swarmanoid," a group of cooperating robots with complementary abilities: "foot-bots" to transport things on the ground, "hand-bots" to climb walls and manipulate

objects, and "eyebots" to fly around, providing information to the other units.

J The military is eager to acquire similar **capabilities**. In one experiment, researchers released a swarm of 66 pint-size robots into an empty office building at Fort A. P. Hill, a training center near Fredericksburg, Virginia. The mission: Find targets hidden in the building.

K Zipping down the main hallway, the 30-centimeter-long (one foot) red robots pivoted this way and that on their three wheels, resembling a group of large insects. Eight sonars³ on each unit helped them avoid collisions with walls and other robots. As they spread out, entering one room after another, each robot searched for objects of interest with a small camera. When one robot encountered another, it used wireless network gear to exchange information. ("Hey, I've already explored that part of the building. Look somewhere else.")

³Sonar is equipment that can detect the position of objects using sound waves.





A huge herd of wildebeest surge across the flooded Mara River in Serengeti National Park, Tanzania.

L In the back of one room, a robot spotted something suspicious: a pink ball in an open closet (the swarm had been trained to look for anything pink). The robot froze, sending an image to its human supervisor. Soon, several more robots arrived to form a perimeter around the pink intruder. Within half an hour, the mission had been **accomplished**—all six of the hidden objects had been found. The research team conducting the experiment declared the run a success. Then they started a new test.

M The demonstration was part of the Centibots project, an investigation to see if as many as a hundred robots could collaborate on a mission. If they could, teams of robots might someday be sent into a hostile village to flush out terrorists or locate prisoners; into an earthquake-damaged building to find victims; onto chemical-spill sites to examine hazardous waste; or along borders to watch for intruders. Military agencies such as DARPA (Defense Advanced Research Projects Agency) have funded a number of robotics programs using collaborative flocks of helicopters and fixed-wing aircraft, schools of torpedo-shaped underwater gliders, and herds of unmanned ground vehicles. But, at the

time, this was the largest swarm of robots ever tested.

N “When we started Centibots, we were all thinking, this is a crazy idea, it’s impossible to do,” says Régis Vincent, a researcher at SRI International in Menlo Park, California. “Now we’re looking to see if we can do it with a thousand robots.”

O In nature, of course, animals travel in even larger numbers. That’s because, as members of a big group, whether it’s a flock, school, or herd, individuals increase their chances of **detecting** predators, finding food, locating a mate, or following a migration route. For these animals, coordinating their movements with one another can be a matter of life or death.

P “It’s much harder for a predator to avoid being spotted by a thousand fish than it is to avoid being spotted by one,” says Daniel Grünbaum, a biologist at the University of Washington. “News that a predator is approaching spreads quickly through a school because fish sense from their neighbors that something’s going on.”

Q When a predator strikes a school of fish, the group is capable of scattering in

patterns that make it almost impossible to track any individual. It might explode in a flash, create a kind of moving bubble around the predator, or fracture into multiple blobs,⁴ before coming back together and swimming away.

R That's the wonderful appeal of swarm intelligence. Whether we're talking about ants, bees, pigeons, or caribou, the ingredients of smart group behavior—decentralized control, response to local cues, simple rules of thumb—add up to a **shrewd** strategy to cope with complexity.

S “We don't even know yet what else we can do with this,” says Eric Bonabeau, a complexity theorist and the chief scientist at Icosystem Corporation in Cambridge, Massachusetts. “We're not used to solving decentralized problems in a decentralized way. We can't control an emergent phenomenon like traffic by putting stop signs and lights everywhere. But the idea of shaping traffic as a self-organizing system, that's very exciting.”

T The internet is already using a form of swarm intelligence. Consider the way Google uses group smarts to find what you're looking for. When you type in a search query, Google surveys billions of Web pages on its index servers⁵ to identify the most relevant ones. One of the ways it ranks the search results is by the number of pages that link to them, counting links as votes (the most popular sites get weighted⁶ votes since they're more likely to be reliable). The pages that receive

more votes are listed higher in the search results. In this way, Google says, it “uses the collective intelligence of the Web to determine a page's importance.”

U Wikipedia, a free collaborative encyclopedia, has also proved to be a big success, with millions of articles in more than 200 languages about everything under the sun, each of which can be contributed by anyone or edited by anyone. “It's now possible for huge numbers of people to think together in ways we never imagined a few decades ago,” says Thomas Malone of MIT's Center for Collective Intelligence. “No single person knows everything that's needed to deal with problems we face as a society, such as health care or climate change, but collectively we know far more than we've been able to tap so far.”

V Such thoughts underline an important truth about collective intelligence: crowds tend to be wise only if individual members act responsibly and make their own decisions. A group won't be smart if its members **imitate** one another, slavishly follow fads, or wait for someone to tell them what to do. When a group is being intelligent, whether it's made up of ants or attorneys, it relies on its members to do their own part. For those of us who sometimes wonder if it's really worth recycling that extra bottle to lighten our impact on the planet, the bottom line is that our actions matter, even if we don't see how.

Adapted from “Swarm Theory,” by Peter Miller: National Geographic Magazine, July 2007

Peter Miller has worked as a reporter for *Life* magazine and as a senior editor at *National Geographic*. He is the author of the best-selling book *The Smart Swarm: How to Work Efficiently, Communicate Effectively, and Make Better Decisions Using the Secrets of Flocks, Schools, and Colonies*.

⁴ A **blob** is an indistinct or shapeless form or object.

⁵ A **server** is a program or device that does a task for other computers on a network, such as store information or run a program.

⁶ If something is **weighted**, it is given more value because it is seen as more important.

UNDERSTANDING THE READING

A UNDERSTAND MAIN IDEAS Note answers to the questions below. Write the paragraph(s) that contain the answers (A–V).

1. What is a “smart swarm”? Explain it in your own words.

_____ Paragraph(s) _____

2. What are the three key aspects of swarm intelligence?

_____ Paragraph(s) _____

3. How does being part of a large group help animals?

_____ Paragraph(s) _____

4. How do search engines and online encyclopedias make use of collective intelligence?

_____ Paragraph(s) _____

B UNDERSTAND PURPOSE Match the paragraphs to their purpose.

- | | |
|-------------------------|---|
| 1. _____ Paragraph B | a. show(s) how we already use swarm intelligence |
| 2. _____ Paragraph G | b. describe(s) potential uses of smart swarm robots |
| 3. _____ Paragraphs H–N | c. explain(s) why individual behavior matters |
| 4. _____ Paragraphs O–Q | d. explain(s) why animals move in swarms |
| 5. _____ Paragraphs T–U | e. list(s) three key features of smart swarms |
| 6. _____ Paragraph V | f. give(s) a general definition of a smart swarm |

C INFER MEANING Find and underline the following phrases in the reading passage. Use the context to identify their meanings. Then write the correct form of each phrase next to its definition.

Critical Thinking

add up to (paragraph A)	set in motion (paragraph D)	blazing the trail (paragraph F)
trial and error (paragraph F)	flush out (paragraph M)	the bottom line (paragraph V)

Review
this Critical
Thinking
Skill in
Unit 4

- _____ to force a person or animal out of hiding
- _____ the essential idea
- _____ to initiate something
- _____ to equal, or result in
- _____ to be the first to do something
- _____ trying different methods until something works